

# Mains electricity and the National Grid

Electricity  
Paper 1 · Combined and Separate  
31 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|    | QUESTION                                                                                 | ANSWER                                                                                      | ✓                        |
|----|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------|
| 1  | What do a.c. and d.c. stand for?                                                         | Alternating current and direct current                                                      | <input type="checkbox"/> |
| 2  | What is an alternating current (a.c.)?                                                   | A current that repeatedly reverses direction.                                               | <input type="checkbox"/> |
| 3  | State the frequency and potential difference of the UK mains electricity supply.         | 50 Hz and about 230 V                                                                       | <input type="checkbox"/> |
| 4  | State the colour of the insulation on the live, neutral and earth wires.                 | Live: brown<br>Neutral: blue<br>Earth: green and yellow stripes                             | <input type="checkbox"/> |
| 5  | State the function of the live wire in a three-core cable.                               | It carries the alternating potential difference from the supply.                            | <input type="checkbox"/> |
| 6  | State the function of the neutral wire in a three-core cable.                            | It completes the circuit.                                                                   | <input type="checkbox"/> |
| 7  | State the function of the earth wire in a three-core cable.                              | It is a safety wire that stops the appliance becoming live.                                 | <input type="checkbox"/> |
| 8  | State the potential difference between the live wire and the earth wire.                 | About 230 V                                                                                 | <input type="checkbox"/> |
| 9  | Name the wires in a three-core cable that are at, or close to, 0 V.                      | The earth wire and the neutral wire                                                         | <input type="checkbox"/> |
| 10 | State when the earth wire will carry a current.                                          | Only when there is a fault.                                                                 | <input type="checkbox"/> |
| 11 | In a plug, name the wire that is connected to the fuse.                                  | The live wire                                                                               | <input type="checkbox"/> |
| 12 | Explain why touching the live wire is dangerous even when the appliance is switched off. | The live wire is still at about 230 V, so a current would flow through the person to earth. | <input type="checkbox"/> |

|    | QUESTION                                                                             | ANSWER                                                                                              | ✓                        |
|----|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------|
| 13 | Define power in terms of energy.                                                     | The rate at which energy is transferred.                                                            | <input type="checkbox"/> |
| 14 | State the standard unit of power.                                                    | Watt (W)                                                                                            | <input type="checkbox"/> |
| 15 | Give two things that affect the energy transferred by an appliance.                  | Its power and the time it is switched on                                                            | <input type="checkbox"/> |
| 16 | What does a higher power rating tell you about an appliance?                         | It transfers more energy each second.                                                               | <input type="checkbox"/> |
| 17 | Describe the National Grid and its purpose.                                          | A system of cables and transformers linking power stations to consumers.                            | <input type="checkbox"/> |
| 18 | State what is transferred across the National Grid.                                  | Electrical power                                                                                    | <input type="checkbox"/> |
| 19 | State the function of transformers.                                                  | To increase or decrease the potential difference.                                                   | <input type="checkbox"/> |
| 20 | What does a step-up transformer do to the potential difference?                      | It increases it.                                                                                    | <input type="checkbox"/> |
| 21 | What does a step-up transformer do to the current?                                   | It decreases it.                                                                                    | <input type="checkbox"/> |
| 22 | Explain why step-up transformers are used before electricity is transmitted.         | A higher potential difference means a smaller current, so less energy is wasted heating the cables. | <input type="checkbox"/> |
| 23 | What is direct current (d.c.)?                                                       | A current that flows in one direction only.                                                         | <input type="checkbox"/> |
| 24 | Name a source of direct current.                                                     | A cell or battery                                                                                   | <input type="checkbox"/> |
| 25 | State the equation linking power, potential difference and current.                  | power = potential difference $\times$ current<br>( $P = V I$ )                                      | <input type="checkbox"/> |
| 26 | State the equation linking power, current and resistance.                            | power = current <sup>2</sup> $\times$ resistance<br>( $P = I^2 R$ )                                 | <input type="checkbox"/> |
| 27 | State the equation linking energy transferred, power and time.                       | energy transferred = power $\times$ time<br>( $E = P t$ )                                           | <input type="checkbox"/> |
| 28 | State the equation linking energy transferred, charge flow and potential difference. | energy transferred = charge flow $\times$ potential difference<br>( $E = Q V$ )                     | <input type="checkbox"/> |

| QUESTION |                                                                                      | ANSWER                                                                                                     | ✓                        |
|----------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------|
| 29       | <b>A battery-powered fan is switched on. Describe the useful energy transfer.</b>    | Energy is transferred electrically from the chemical store of the battery to the kinetic store of the fan. | <input type="checkbox"/> |
| 30       | <b>Explain why step-down transformers are used before electricity reaches homes.</b> | To reduce the potential difference to a safer value (230 V).                                               | <input type="checkbox"/> |
| 31       | <b>Explain why any connection between the live wire and earth is dangerous.</b>      | A very large current would flow, which could cause a fire or a fatal shock.                                | <input type="checkbox"/> |